

08 Geometric Transformations

8.6 Reflection

Building a reflection matrix from a normal vector — symmetric, orthogonal, self-inverse, determinant -1 , and its Householder link to projection



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Reflection maps a point to its mirror position across an axis or plane of symmetry

3

A reflection matrix $\mathbf{M}$ is symmetric, and it is also orthogonal

5

Reflection preserves area but flips orientation, so the determinant flips sign: $\det(\mathbf{M}) = -1$

7

This construction works in 2D, 3D, and any n -dimensional Euclidean space

2

Build a reflection matrix from a normal vector $\mathbf{n}$:
 $\mathbf{M} = \mathbf{I} - 2\hat{\mathbf{n}}\hat{\mathbf{n}}^T$

4

A reflection matrix satisfies $\mathbf{M}^2 = \mathbf{I}$ — its inverse is itself

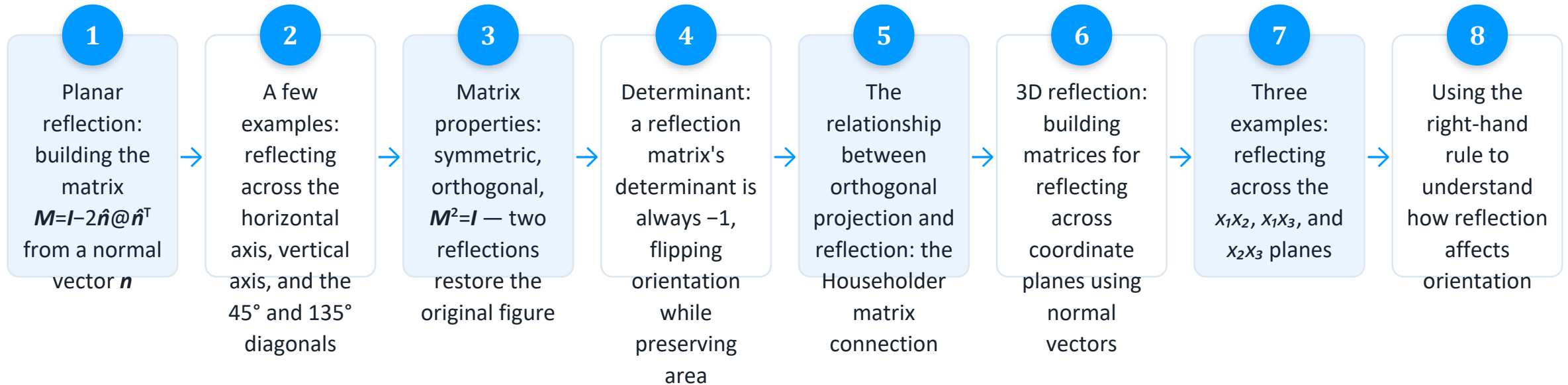
6

The relationship between projection matrices and reflection matrices involves the Householder matrix

8

In 3D, build reflection matrices onto coordinate planes using normal vectors

From Building a Reflection Matrix to 3D Reflection



Seven Lenses for Every Transform in This Chapter

- 1 Matrix-multiplication view: how the matrix acts on $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_2$, and the all-ones vector $\mathbf{1}$
- 2 The determinant: $|\det|$ greater than 1? Positive, negative, or zero? Determinants of special matrices
- 3 How the determinant reflects the area or volume change caused by the transform
- 4 The inverse (does it exist?) and the inverse transform; inverses of special matrices
- 5 Linear-combination view: parallelogram/parallelepiped grids, and whether dimension drops
- 6 Does the transform matrix form a basis, and what kind of basis?
- 7 Properties of the transform matrix: symmetric, diagonal, orthogonal, invertible, commutative

Reflecting across the line through the origin with normal n , the reflection matrix is M

$$M = I - 2(n @ n^T) / \|n\|^2 \quad M = I - 2\hat{n} @ \hat{n}^T$$

- Reflection maps a point to its mirror position across some axis or plane of symmetry
- In particular, when the normal is already a unit vector $\hat{n}$, this simplifies to $M = I - 2\hat{n} @ \hat{n}^T$

The normal vector n of the line $a_1x_1+a_2x_2=0$ (through the origin)

$$n = [a_1, a_2]^T$$

- Pick one unit normal vector, and you can build the reflection transform for the hyperplane it defines
- This normal-vector construction works not only in 2D and 3D, but in any n-dimensional Euclidean space

Substituting $n=[a_1,a_2]$ into $M=I-2\hat{n}@\hat{n}$ expands to

$$M = (1 / (a_1^2 + a_2^2)) [a_2^2 - a_1^2, -2a_1a_2; -2a_1a_2, a_1^2 - a_2^2]$$

- This expansion closely parallels the projection-matrix derivation, with an extra factor of 2
- Let's build intuition for reflection with a few concrete examples

Reflecting across the horizontal axis ($x_2=0$), unit normal $\hat{n}=[0,1]$

$$M = I - 2\hat{n}\hat{n}^T = [1, 0; 0, -1]$$

- $[x_1, x_2]^T$ reflected across the horizontal axis becomes $[x_1, -x_2]^T$: x_1 unchanged, x_2 flips sign

Figure 1 · Reflecting Across the Horizontal and Vertical Axes

- (a) Reflecting across the horizontal axis: x_1 unchanged, x_2 flips sign; (b) across the vertical axis: x_1 flips sign, x_2 unchanged

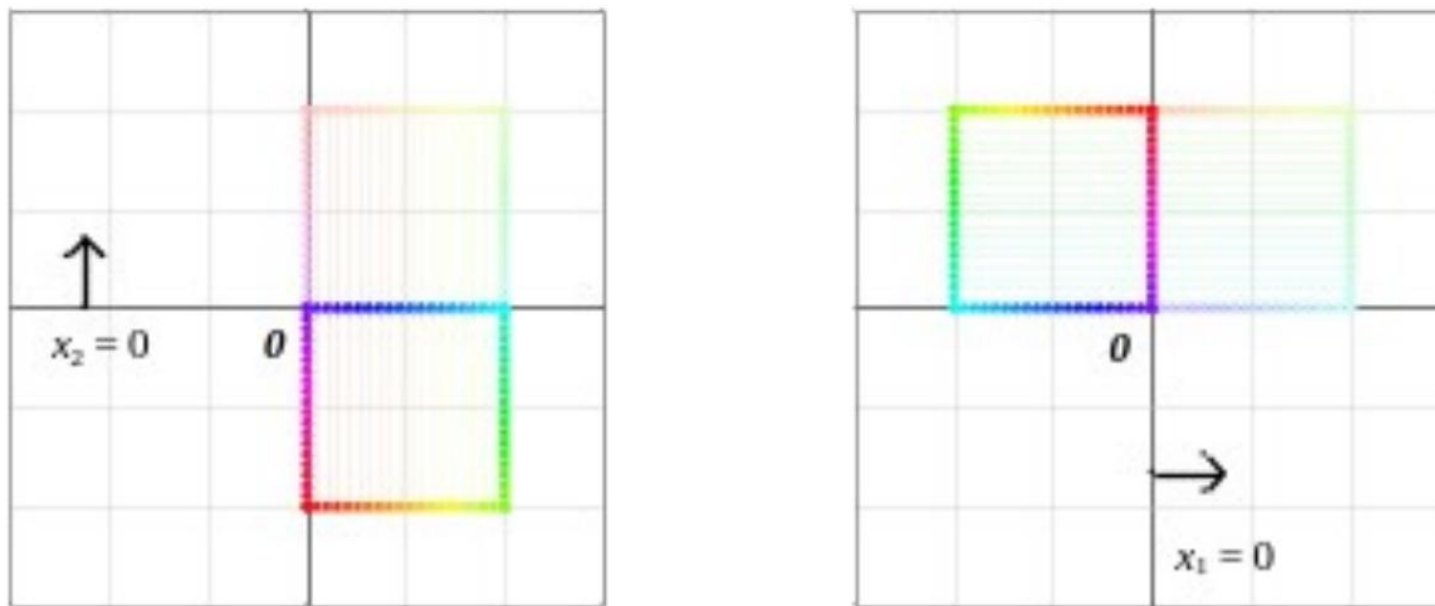


图 1. 关于横轴、纵轴对称

Reflecting across the vertical axis ($x_1=0$), unit normal $\hat{n}=[1,0]$

$$M = I - 2\hat{n}\hat{n}^T = [-1, 0; 0, 1]$$

- $[x_1, x_2]^T$ reflected across the vertical axis becomes $[-x_1, x_2]^T$: x_1 flips sign, x_2 unchanged

Reflection matrices for $x_1 - x_2 = 0$ and $x_1 + x_2 = 0$

$$M(x_1 - x_2 = 0) = [0, 1; 1, 0] \quad M(x_1 + x_2 = 0) = [0, -1; -1, 0]$$

- Reflecting across $x_1 - x_2 = 0$ swaps the two coordinates: $[x_1, x_2]^T \rightarrow [x_2, x_1]^T$
- Reflecting across $x_1 + x_2 = 0$ swaps the two coordinates and negates them: $[x_1, x_2]^T \rightarrow [-x_2, -x_1]^T$

Figure 2 · Reflecting Across the 45° and 135° Diagonals Through the Origin

- (a) Reflecting across $x_1 - x_2 = 0$; (b) reflecting across $x_1 + x_2 = 0$ — the two axes of reflection are perpendicular

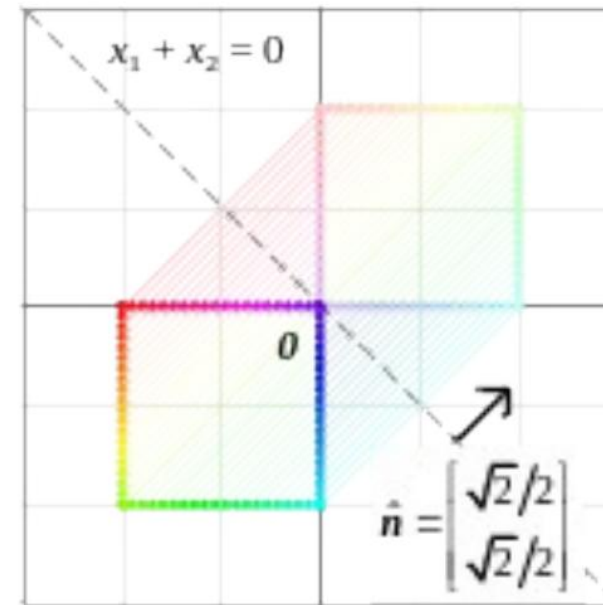
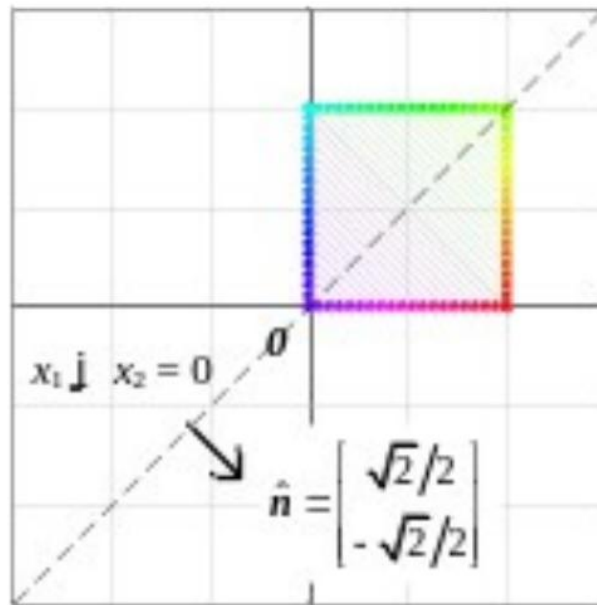


图 2. 关于过原点 45 度、135 度斜线对称

The reflection matrix M is symmetric

$$M = M^T$$

- This is visible directly from the expanded form of M — it is symmetric about the main diagonal
- This property holds in any dimension, for any normal vector

The reflection matrix M is also orthogonal

$$M @ M^T = M^T @ M = I$$

- This means the inverse of a reflection is the reflection itself
- It also means the columns of M form an orthonormal basis

Combining symmetric and orthogonal, a reflection matrix satisfies this identity

$$M^2 = I$$

- This means applying the same reflection twice restores the original figure
- Geometrically, reflection is its own inverse transform — a property that sets it apart from rotation and scaling

Figure 3 · Applying the Same Reflection Twice Restores the Original Figure

- The original shape changes after one reflection M , then a second reflection ($M^2=I$) restores it exactly

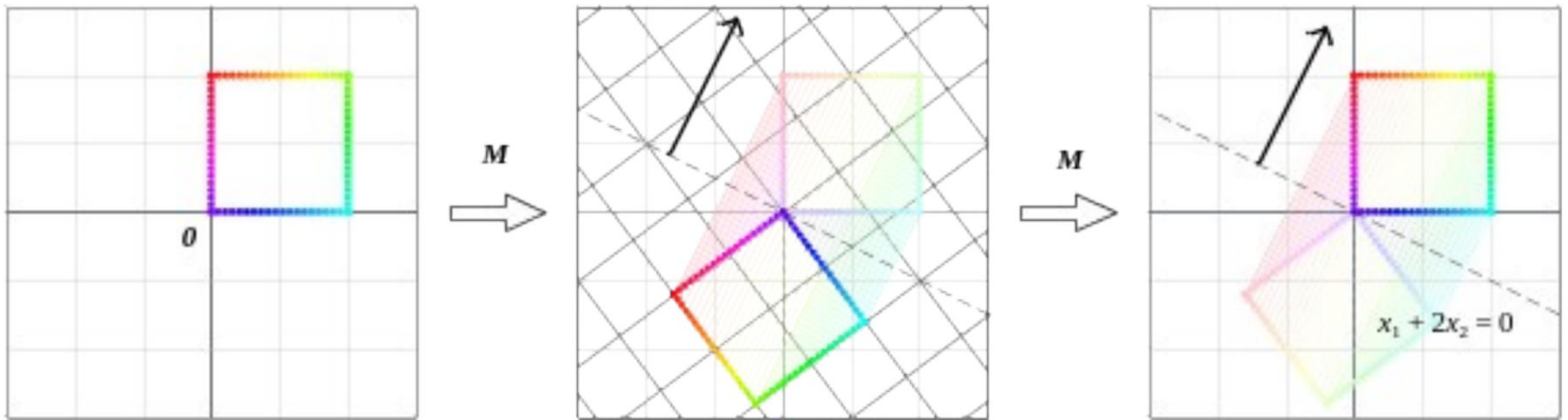


图 3. 同一个镜像连续操作两次，还原原始图形

A planar reflection matrix always has determinant -1

$$\det(M) = -1$$

- A determinant's sign relates to the orientation of a transform: positive preserves the order of basis vectors, negative reverses it
- Reflection preserves area but flips the image, swapping the order of the column vectors — the key feature distinguishing it from rotation (determinant 1)

The matrix $M=I-2\hat{n}\hat{n}^T$ in this formula has a name — the Householder matrix

$$\mathbf{a} - \mathbf{b} = 2 \text{proj}_{\mathbf{n}}(\mathbf{a})$$

- Vector $\mathbf{a}$ and vector $\mathbf{b}$ are reflections of each other across the line through the origin with normal $\mathbf{n}$
- Their difference is exactly twice the projection of $\mathbf{a}$ onto $\mathbf{n}$ — revealing a tight link between reflection and orthogonal projection

Figure 4 · The Householder Matrix and Orthogonal Projection

- Subtracting $\mathbf{b}$ (the reflection of $\mathbf{a}$ across $\mathbf{n}$) from $\mathbf{a}$ gives exactly twice $\text{proj}_{\mathbf{n}}(\mathbf{a})$

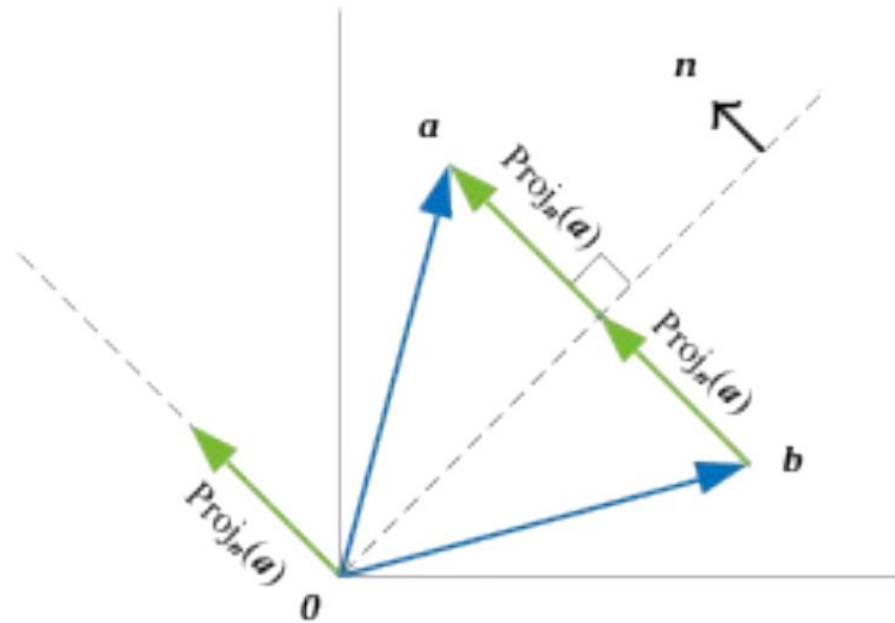


图 4. 豪斯霍尔矩阵和正交投影

This same reflection matrix works in 3D and higher dimensions

$$M = I - 2\hat{n}\hat{n}^T$$

- For a plane, its normal vector is simply the vector perpendicular to it
- All you need is the unit normal $\hat{n}$ of the hyperplane to build the corresponding reflection matrix

Figure 5 · Normal Vectors of Planes, in 3D Space

- (a) The x_1x_2 plane has normal $[0,0,1]^T$; (b) the x_1x_3 plane has normal $[0,1,0]^T$; (c) the x_2x_3 plane has normal $[1,0,0]^T$

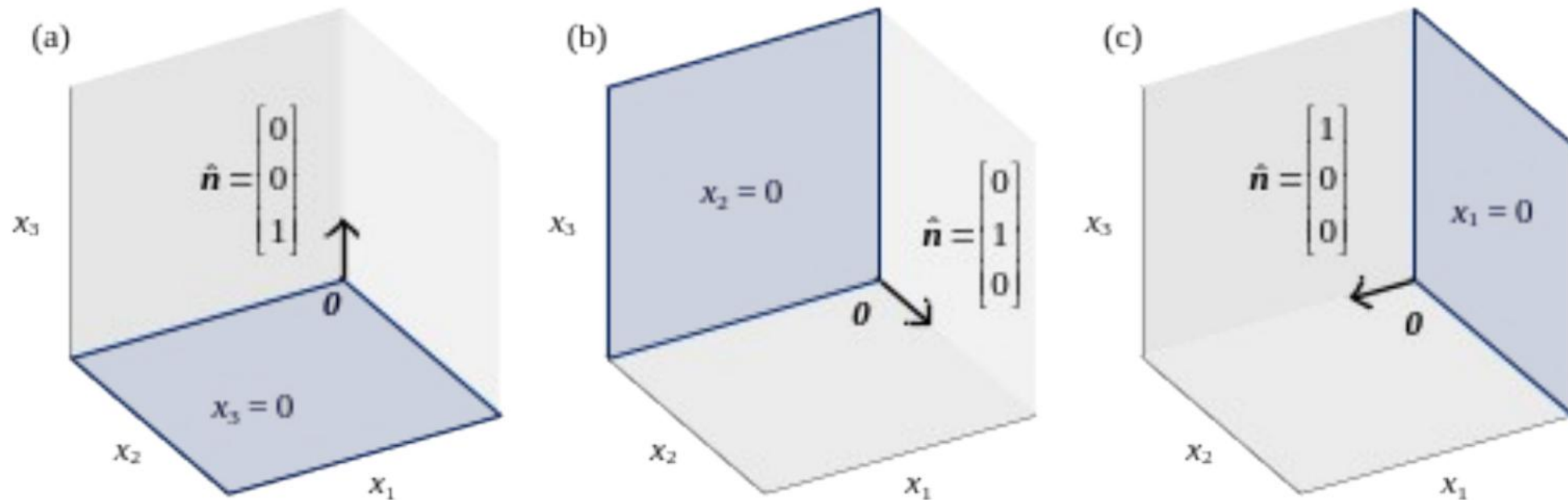


图 5. 平面的法向量

The unit normal of the x_1x_2 plane is $[0,0,1]$; its reflection matrix is

$$\mathbf{M} = \text{diag}(1, 1, -1)$$

- A 3D column vector $\mathbf{x}$ reflected across the x_1x_2 plane: x_1 and x_2 unchanged, x_3 flips sign
- Try computing this reflection matrix's determinant yourself

Figure 6 · Reflecting Across the x_1x_2 Plane, in 3D

- The unit cube, reflected across the x_1x_2 plane, flips its edges along x_3 to the opposite side

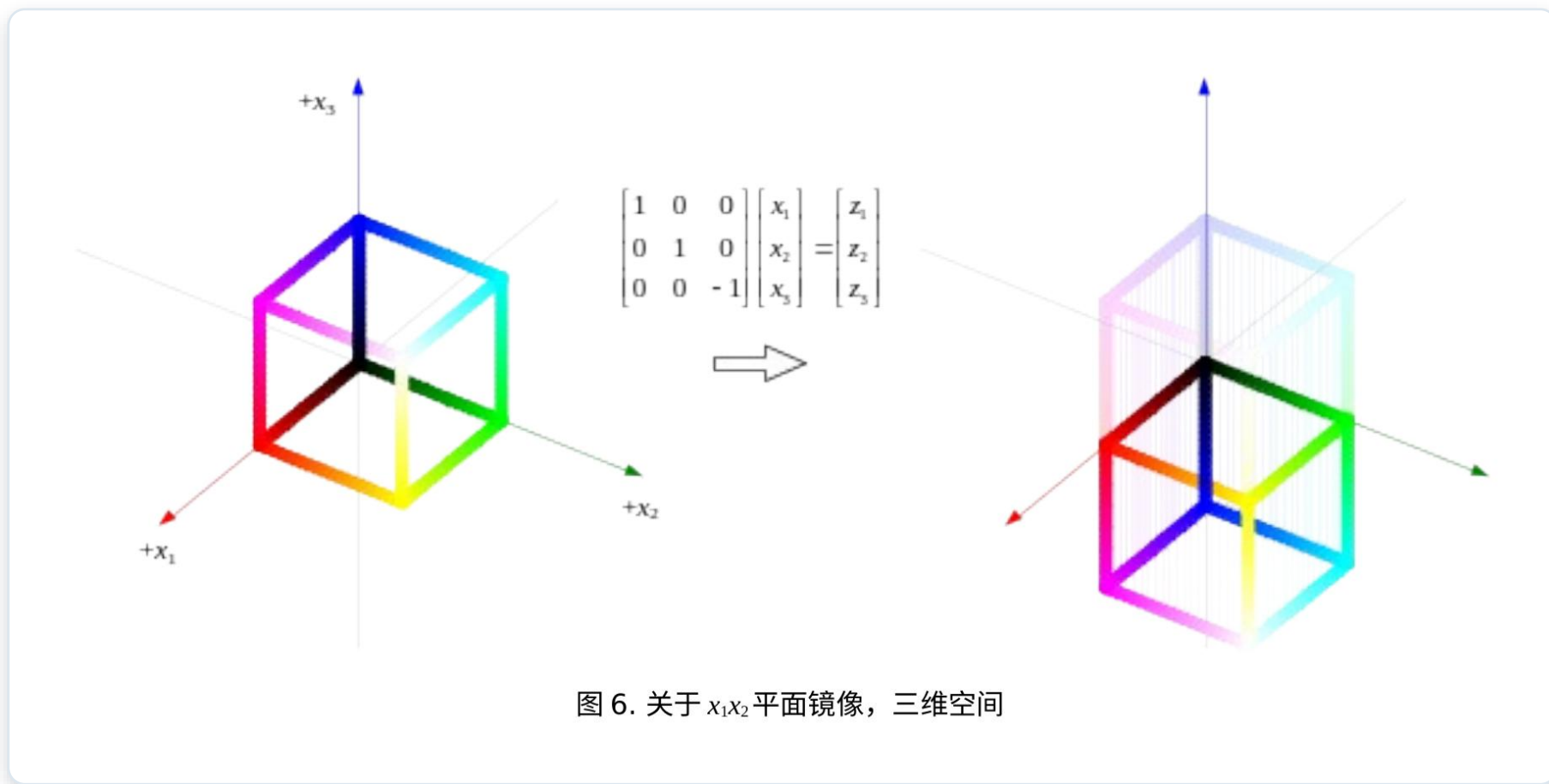
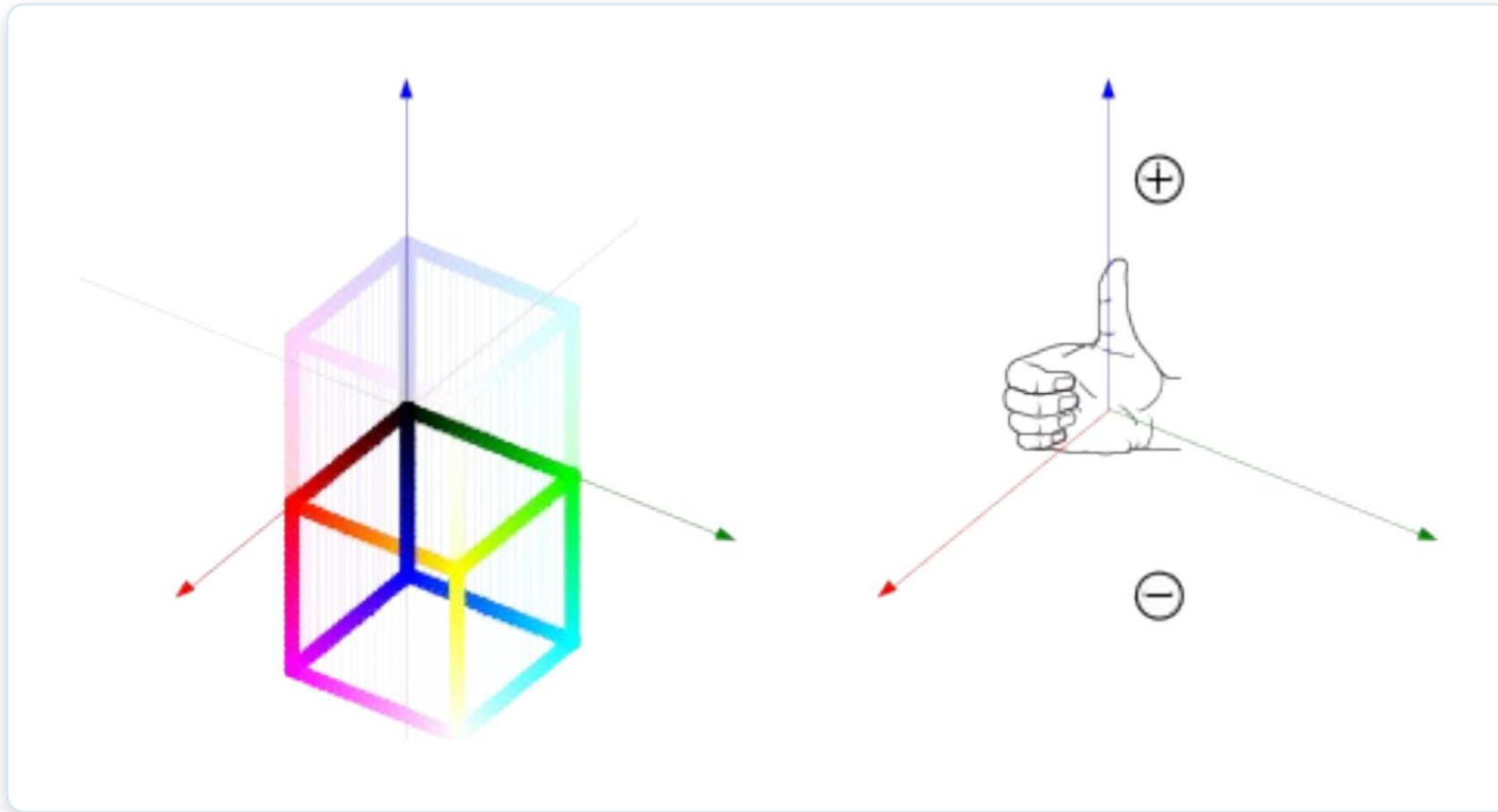


Figure 7 · Using the Right-Hand Rule to Judge Determinant Sign

- A reflection's determinant is negative, flipping the orientation of the coordinate system — the right-hand rule makes this change visible



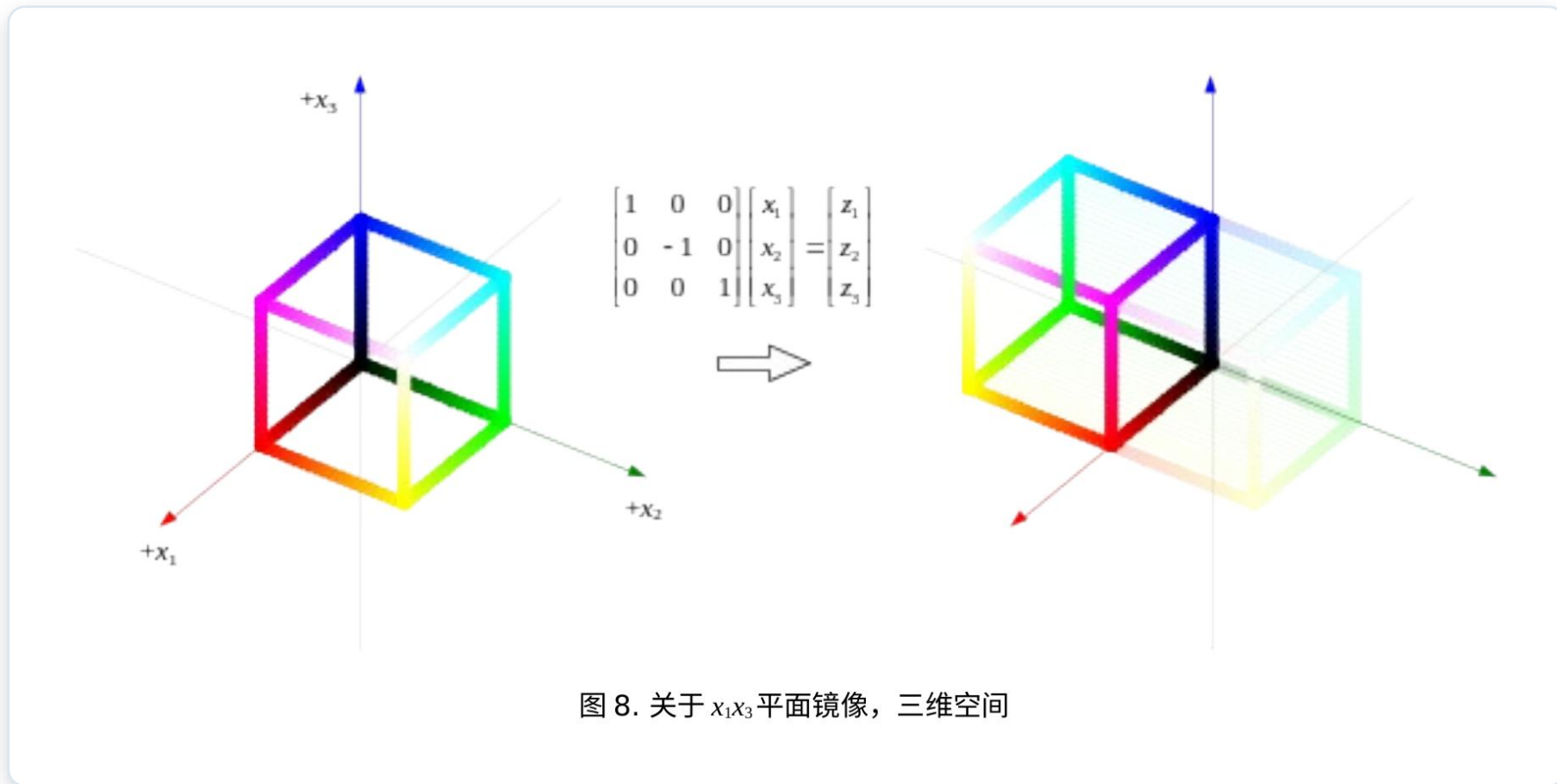
The unit normal of the x_1x_3 plane is $[0,1,0]$; its reflection matrix is

$$\mathbf{M} = \text{diag}(1, -1, 1)$$

- A 3D column vector $\mathbf{x}$ reflected across the x_1x_3 plane: x_1 and x_3 unchanged, x_2 flips sign

Figure 8 · Reflecting Across the x_1x_3 Plane, in 3D

- The unit cube, reflected across the x_1x_3 plane, flips its edges along x_2 to the opposite side



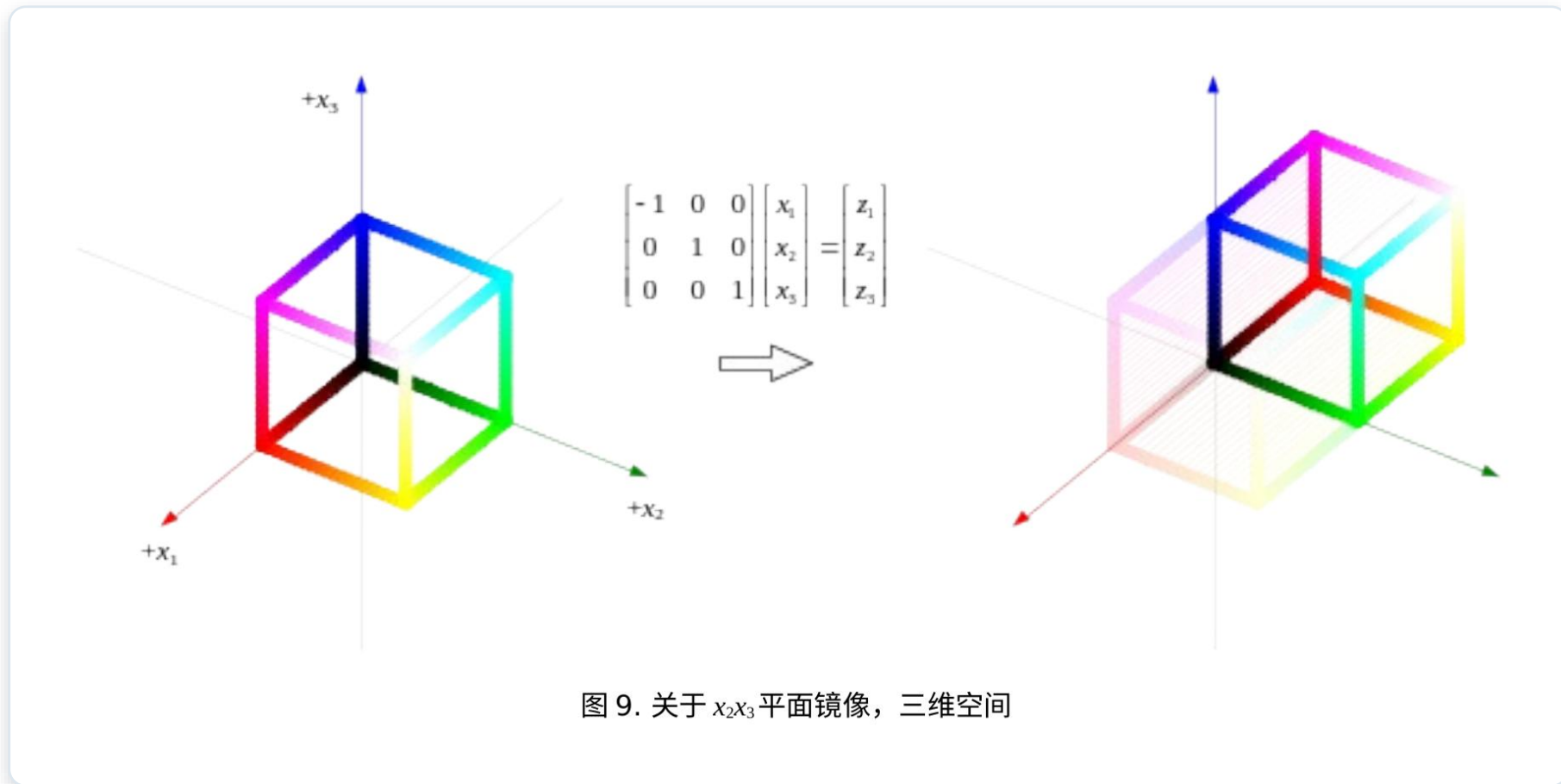
Try computing the reflection matrix for reflecting across the x_2x_3 plane

$$M = \text{diag}(-1, 1, 1)$$

- The unit normal of the x_2x_3 plane is $[1,0,0]$; after reflection x_1 flips sign, x_2 and x_3 are unchanged

Figure 9 · Reflecting Across the x_2x_3 Plane, in 3D

- The unit cube, reflected across the x_2x_3 plane, flips its edges along x_1 to the opposite side



SUMMARY

What Have We Seen from the Matrix View of Reflection?

A reflection matrix is built from a unit normal: $\mathbf{M} = \mathbf{I} - 2\hat{\mathbf{n}}\hat{\mathbf{n}}^T$ — the Householder matrix

$\mathbf{M}$ is both symmetric and orthogonal, and satisfies $\mathbf{M}^2 = \mathbf{I}$ — reflection is its own inverse

A reflection matrix's determinant is always -1 : area is preserved, but orientation flips and column order swaps

Reflection is tightly linked to orthogonal projection: $\mathbf{a} - \mathbf{b} = 2\text{proj}_{\mathbf{n}}(\mathbf{a})$, i.e. $\mathbf{M} = \mathbf{I} - 2\mathbf{P}$

Glossary

Reflection 镜像 —— a geometric transform mapping a point to its mirror position across an axis or plane

Normal Vector 法向量 —— n , the vector perpendicular to the axis or plane of reflection

Symmetric Matrix 对称矩阵 —— a matrix satisfying $M=M^T$

Involutory Matrix 对合矩阵 —— a matrix satisfying $M^2=I$, i.e. its own inverse

Orthogonal Projection 正交投影 —— $P=\hat{n}\hat{n}^T$, related to reflection via $M=I-2P$

Householder Matrix 豪斯霍尔德矩阵 —— $M=I-2\hat{n}\hat{n}^T$, a reflection matrix built from a unit normal

Reflection Matrix 镜像矩阵 —— M , the matrix that performs a reflection transform

Orthogonal Matrix 正交矩阵 —— a matrix satisfying $M @ M^T = I$

Determinant Sign Flip 行列式变号 —— a reflection matrix's determinant is always -1

Orthonormal Basis 规范正交基 —— the basis formed by the columns of a reflection matrix M

EXPLORE FURTHER

Further Reading

Recap · 8.5 Orthogonal Projection

Earlier in this chapter — how projection matrices are built and their properties

Further Reading · 9.1 Orthogonality

Next chapter — orthogonal vectors, orthogonal matrices, and orthonormal bases in full

Further Reading · 9.4 Gram–Schmidt Orthogonalization

Next chapter — building orthogonal bases using projection and reflection ideas

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q9

Q1 For lines through the origin such as $-x_2=0$, $-x_1=0$, $x_1-2x_2=0$, compute the normal vector and reflection matrix.

Q3 Modify LA_08_02_01.ipynb to visualize the planar reflections in Figures 1 and 2.

Q5 For hyperplane normal vectors $[0,0,0,1]^T$, $[0,0,1,0]^T$, $[1,1,1,1]^T$, compute the reflection matrices.

Q7 Prove that a reflection matrix M satisfies M equals M transposed, and M times M equals the identity I .

Q9 Write a Python function `reflect_matrix(n)` that returns the reflection matrix for a given normal vector.

Q2 For 3D planes through the origin such as $x_2=0$, $x_1=0$, $x_1+x_2=0$, compute the normal vector and reflection matrix.

Q4 Modify LA_08_03_01.ipynb to visualize the 3D reflections in Figures 6, 8, and 9.

Q6 For hyperplanes $x_1+x_2+x_3+x_4=0$ and $x_1-x_2-x_3+x_4=0$, compute the reflection matrices.

Q8 Use numpy to verify: every reflection matrix M in this section has determinant -1 .

Q10 – Q17

Q10 Use matplotlib to visualize an arbitrary vector $\mathbf{a}$ and its reflection across the line $x_1 + 2x_2 = 0$.

Q12 Write a Python function `is_reflection_matrix(M)` that checks whether a matrix is symmetric, orthogonal, and has determinant -1 .

Q14 Explore the algebraic relationship between the Householder matrix $M = I - 2\hat{\mathbf{n}}\hat{\mathbf{n}}^T$ and the projection matrix $P = \hat{\mathbf{n}}\hat{\mathbf{n}}^T$; verify with code that $M = I - 2P$.

Q16 Use `numpy.linalg.eig` to compute the eigenvalues of a reflection matrix M , and observe they can only be 1 or -1 .

Q11 Explore: a reflection matrix M is its own inverse — verify with code that $M @ M$ equals the identity.

Q13 Use Python to compare reflecting a batch of points across one line via matrix multiplication versus point-by-point computation.

Q15 Given the plane normal vector $\mathbf{n} = [1, 2, 2]/3$, compute the corresponding reflection matrix and verify it is orthogonal.

Q17 Comprehensive: for the 4D hyperplane $x_1 + x_2 + x_3 + x_4 = 0$, compute the reflection matrix in Python and verify $M @ M$ equals I .

Q18 – Q25

Q18 Explore: is the composition of two different reflections (first across n_1 , then across n_2) still a reflection? Verify with code and check the determinant.

Q20 Explore: in 3D, reflecting across two different coordinate planes in succession (first x_1x_2 , then x_1x_3) corresponds to what geometric operation?

Q22 Write a Python function `householder_matrix(v)` that returns the Householder matrix for an arbitrary nonzero normal vector v .

Q24 Think through and compute: can the identity matrix I be viewed as a special kind of reflection? What is its determinant?

Q19 Write code comparing: projecting a square then reflecting it, versus reflecting it directly — how do the results differ?

Q21 Use the right-hand rule to judge: after reflecting across the x_2x_3 plane, does the coordinate system still satisfy the right-hand rule? Verify the determinant's sign with code.

Q23 Explore: reflection matrices and rotation matrices are both orthogonal, but their determinant signs differ — verify with code and summarize the geometric difference.

Q25 Preview thought: the next chapter covers orthogonality (9.1) — how does this section's orthogonal matrix M relate to Gram matrices and orthonormal bases?

10 Prompts to Explore With AI

1 Explain why a reflection matrix M can be constructed from a unit normal vector $\hat{n}$ as $M = I - 2\hat{n}\hat{n}^T$.

2 Prove that a reflection matrix M satisfies M equals M transposed, and M times M equals the identity.

3 Explain why a reflection matrix's determinant is always -1 , and how that differs from a rotation matrix's determinant of 1 .

4 Using Python, demonstrate how reflecting a shape twice across the same line restores its original form.

5 Explain the relationship between the Householder matrix $M = I - 2\hat{n}\hat{n}^T$ and the projection matrix $P = \hat{n}\hat{n}^T$.

10 Prompts to Explore With AI

6 Explain why reflection preserves a figure's area but changes the order of its basis vectors.

7 Use Python to verify that matrices reflecting across the 3D coordinate planes are always diagonal, with entries ± 1 .

8 Explain how the right-hand rule helps judge the change in a coordinate system's orientation before and after reflection.

9 Explain why reflection matrices and rotation matrices are both orthogonal, yet geometrically completely different.

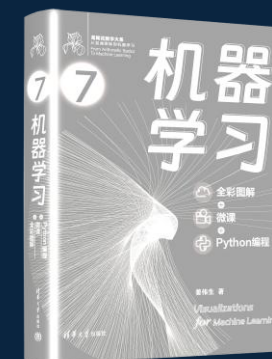
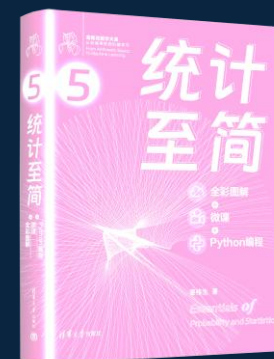
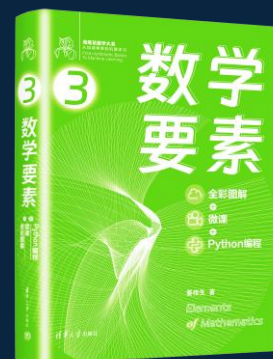
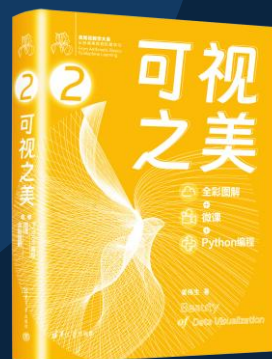
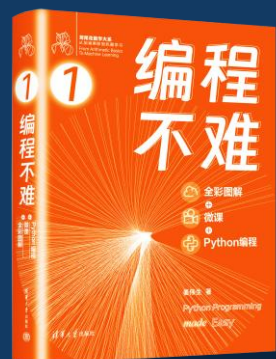
10 Explain what applications reflection transforms might have in computer graphics or the analysis of physical symmetry.

数学

数学基础

线性代数

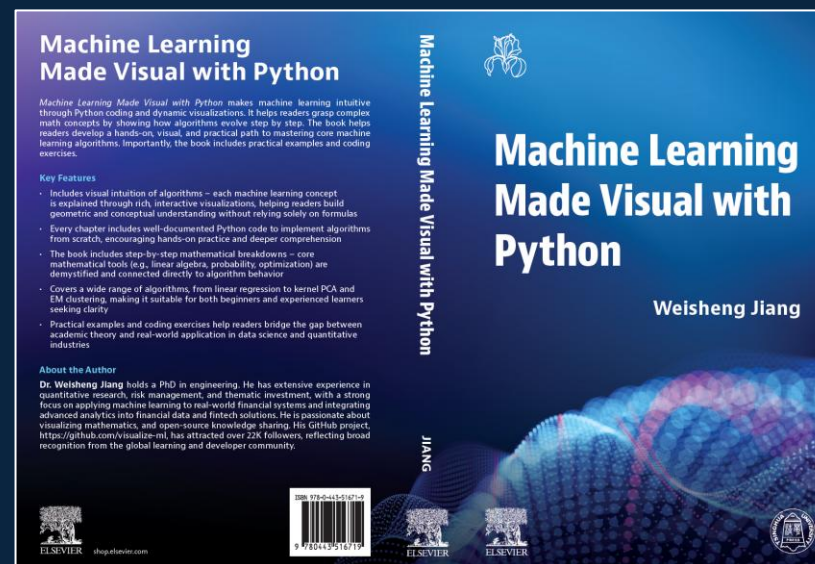
概率统计



Python编程

数据可视化

工具



数据分析
图和网络

回归、降维
分类、聚类

实践